Reg. No. | | | | | | | | | | | [14EC663]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

DIGITAL SWITCHING SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least TWO from each part
2. Missing data may be suitably assumed.

PART – A

1.	**a.**	Explain the operation of four-wire circuit used in the two-way transmission system and derive the expression for stability margin 'M'.	**08**
	b.	Explain neatly with diagram, the evolution of message switching system.	**06**
2.	**a.**	Explain basic central office linkage with neat diagram.	**09**
	b.	Explain basic control structure of a central office.	**05**
3.	**a.**	Derive the expression for iterative form of Erlang's lost call formula.	**10**
	b.	Define the following.	**04**
		i. Erlang	
		ii. Congestion	
4.	**a.**	Design a progressive grading to connect 20 outgoing trunks to switches of availability 10.	**08**
	b.	Design a two stage switching network for connecting 200 incoming trunks to 200 outgoing trunks.	**06**

PART – B

5.	**a.**	Explain frame alignment of PCM signals entering a digital exchange.	**06**
	b	With neat diagram explain space and time switching.	**08**
6.	**a.**	With neat diagram explain basic software architecture of a typical digital switching system.	**08**
	b.	With neat block diagram, explain software linkage during call.	**06**
7.	**a.**	Write a short note on detect analysis.	**06**
	b.	Explain briefly with neat block diagram organizational interfaces of a typical digital switching system central office.	**08**
8.	**a.**	Explain three level scheme of recovery strategy in digital switch.	**06**
	b.	Explain generic switch software architecture.	**08**

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